INTERNSHIP MINI PROJECTS

Traffic Data Analyzer

Description:

Analyze traffic flow data.

Features:

- Peak hour identification
- Average vehicle count
- Weekly comparison
- Line graph

MINI PROJECT : TRAFFIC DATA ANALYZER

Abstract

The Traffic Data Analyzer project is a Python-based application designed to analyze and visualize traffic flow data. The system provides user registration and login functionality using a MySQL database. After successful login, users can access a dashboard that analyzes traffic data stored in a CSV dataset. The system identifies peak traffic hours, calculates average vehicle count, compares weekly traffic data, and displays graphical visualizations using Matplotlib. The application is developed using Python with Tkinter for the graphical user interface and Pandas for data analysis.

Introduction

Traffic congestion is a common problem in urban areas. Analyzing traffic patterns helps in understanding peak traffic times and improving transportation planning. The Traffic Data Analyzer system provides tools to analyze traffic flow data using Python-based technologies. The application allows users to register, log in, and analyze traffic data through an interactive dashboard.

Objectives

- To analyze traffic flow data using Python
- To identify peak traffic hours
- To calculate average vehicle counts
- To compare weekly traffic data
- To visualize traffic patterns using graphs
- To provide a user-friendly dashboard

Technologies Used

- Python : Programming language used to build the application.
- Tkinter : Used to create the graphical user interface.
- Pandas : Used to analyze the CSV traffic dataset.
- NumPy : Used for numerical operations.
- Matplotlib : Used for generating graphs.
- MySQL : Used for storing registered user details.
- CSV Dataset : Used for storing traffic flow data.

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System Architecture

Loading Screen

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Registration Screen

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Login Screen

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User Dashboard

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Traffic Data Analysis

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Graph Visualization

Dataset Description

The dataset used in this project contains traffic flow information including time, vehicle count, and day of the week. The dataset contains more than 100 rows of traffic data which is analyzed using Pandas.

Example data:

time,vehicle_count,day

09:00,150,Monday

10:00,180,Monday

11:00,200,Tuesday

Features of the System

• Loading screen

• User registration

• User login authentication

• Traffic data analysis

• Peak hour identification

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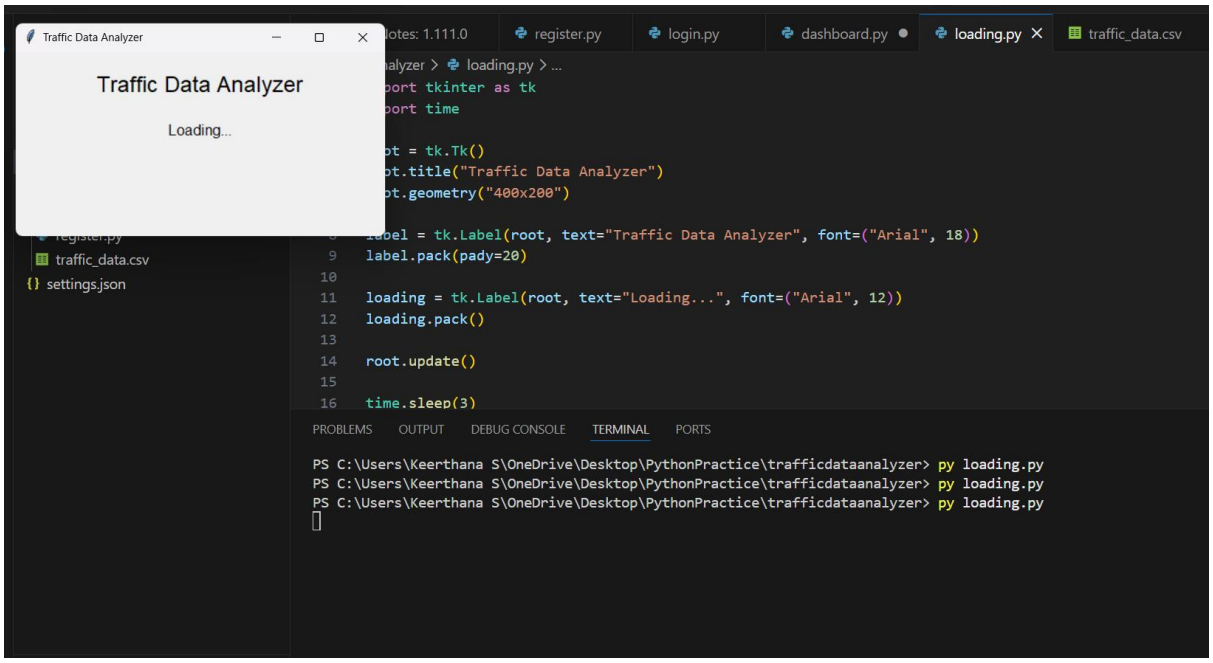
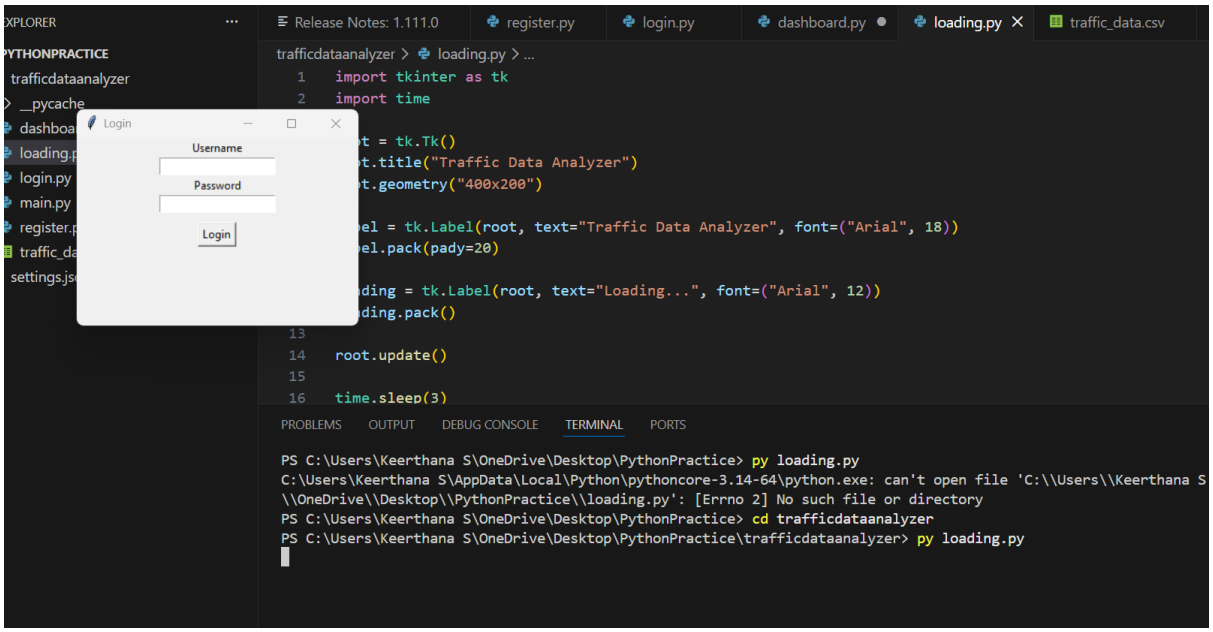
- Average vehicle count calculation
- Weekly comparison
- Line graph visualization

Screenshots of project





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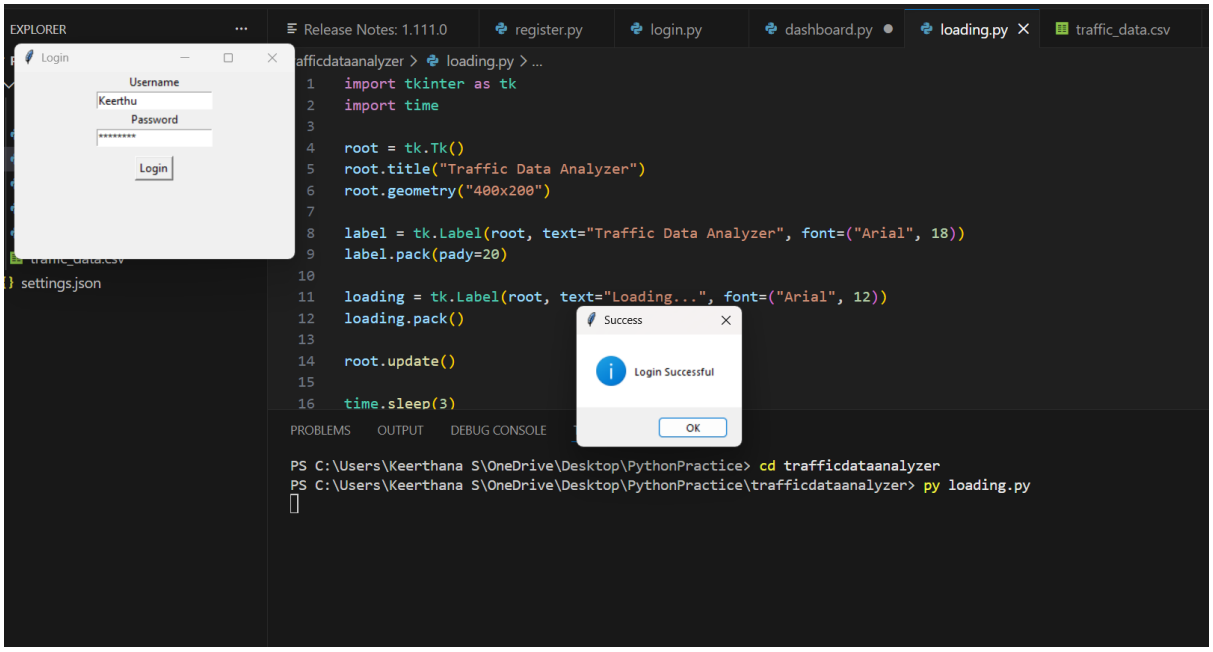

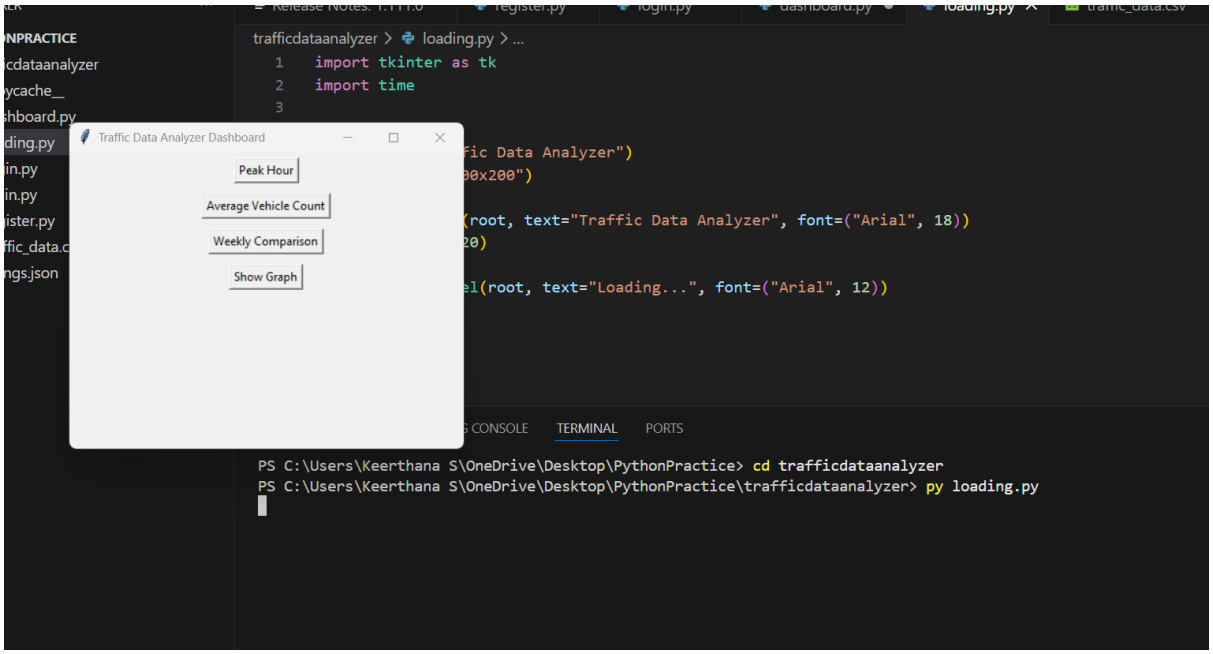

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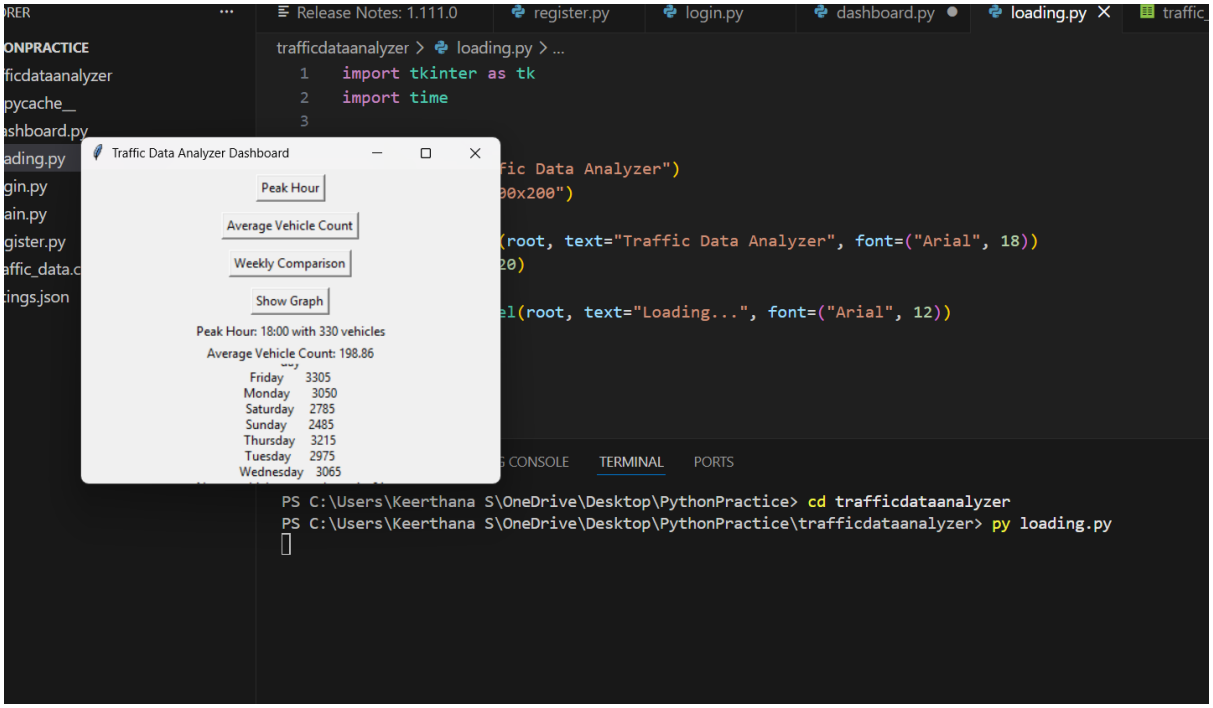

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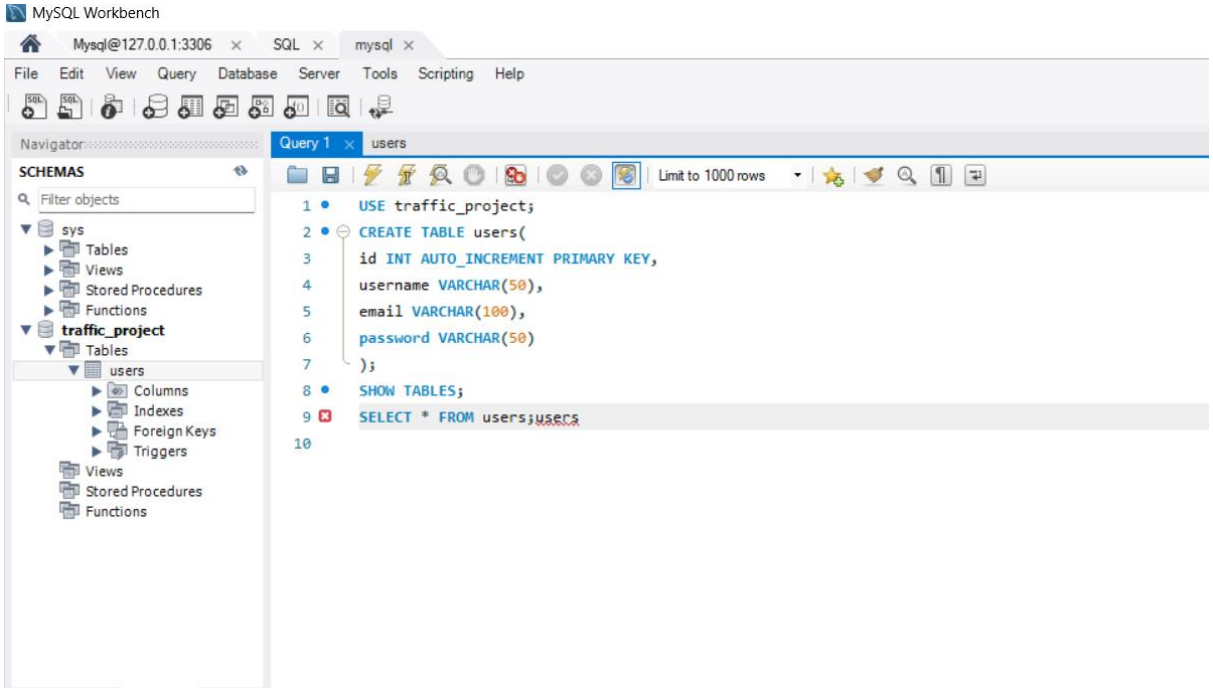

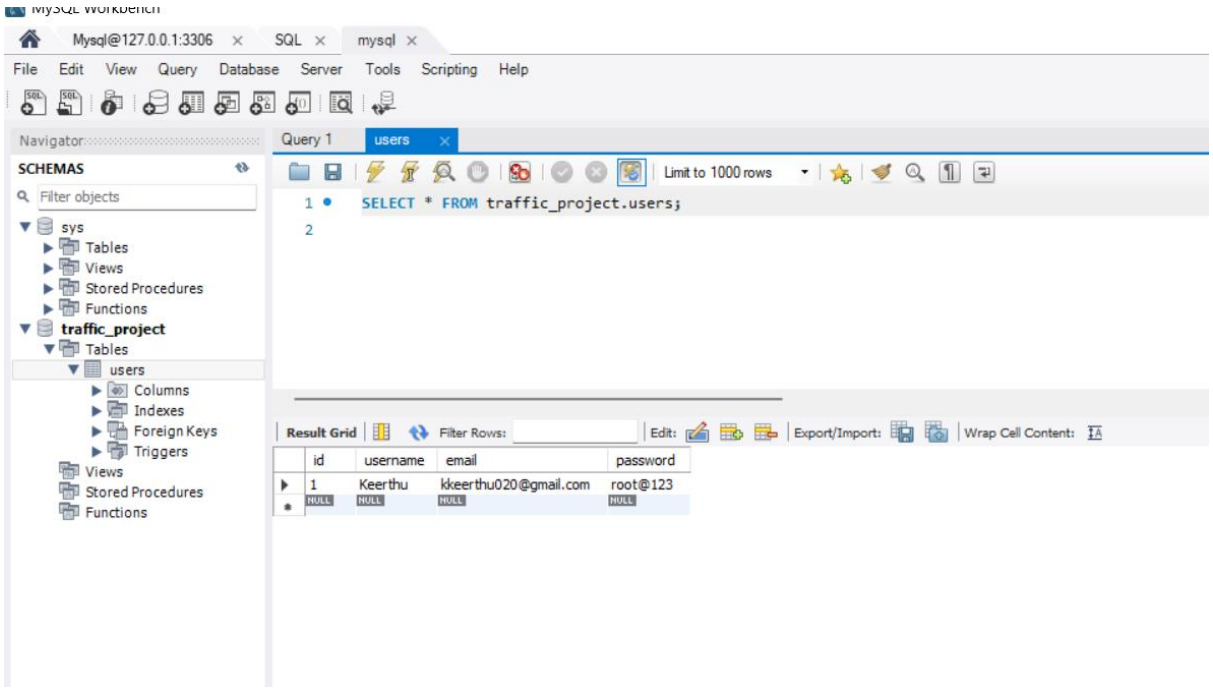

Conclusion

The Traffic Data Analyzer project successfully demonstrates how Python can be used for data analysis and visualization. The system allows users to analyze traffic patterns and identify peak traffic hours. The project combines database management, graphical user interfaces, and data visualization techniques.